

Credit Card Services

Business Analysis Report

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Executive Summary

This report presents a detailed analysis of customer behavior in a credit card dataset using clustering techniques. The primary objective was to segment customers into meaningful groups that reveal unique spending patterns, interaction levels with the bank, and credit card usage behavior.

The study was carried out in the following steps:

- **Exploratory Data Analysis (EDA):** Understanding distributions, relationships, and patterns.
- **Data Preprocessing:** Ensuring the dataset was cleaned, scaled, and prepared for modeling.
- **K-Means Clustering:** Implemented and evaluated using the elbow method and silhouette scores.
- **Hierarchical Clustering:** Tested with multiple linkage methods, evaluated using cophenetic correlation, and compared with K-Means.
- **Cluster Profiling:** Summarized the distinct characteristics of each cluster.
- **Business Insights and Recommendations:** Actionable strategies derived from the findings.

Key findings include:

- **Three main customer groups were identified.**
 - A **high-value, digital-savvy group** with high credit limits and online engagement.
 - A **mass-market group** with moderate credit use and mixed bank interactions.
 - A **budget-conscious group** with low credit limits and frequent offline interactions.
- K-Means and Hierarchical clustering largely agreed, strengthening confidence in the identified segments.
- Recommendations include **personalized credit offers, digital engagement campaigns, and targeted customer retention strategies.**

Introduction

The credit card industry is highly competitive, and companies must understand their customer base in detail to design effective marketing strategies, improve customer experience, and reduce costs.

Currently, the company has customers with very different behaviors:

- Some use credit cards heavily and prefer digital channels.
- Others have low credit limits and rely more on physical branches or phone support.
- A few have high limits but do not engage much at all.

The **objective** of this project is to **segment customers into meaningful groups (clusters)** using clustering techniques. These clusters will help the company to:

- Personalize services and offers.
- Identify high-value vs low-value customers.
- Understand channel preferences (online vs bank/calls).
- Improve efficiency by targeting the right customer with the right product.

This will be done using **K-means Clustering** and **Hierarchical Clustering**, and results will be compared.

Exploratory Data Analysis (EDA)

Univariate Analysis

Avg_Credit_Limit:

- Ranges from very small (~10,000) to very large (100,000+).
- Suggests the company has both mass-market and elite customers.

Total_Credit_Cards:

- Most people have between 2 and 8 credit cards.
- A small group holds more than 10, which may indicate premium or heavy users.

Total_visits_bank:

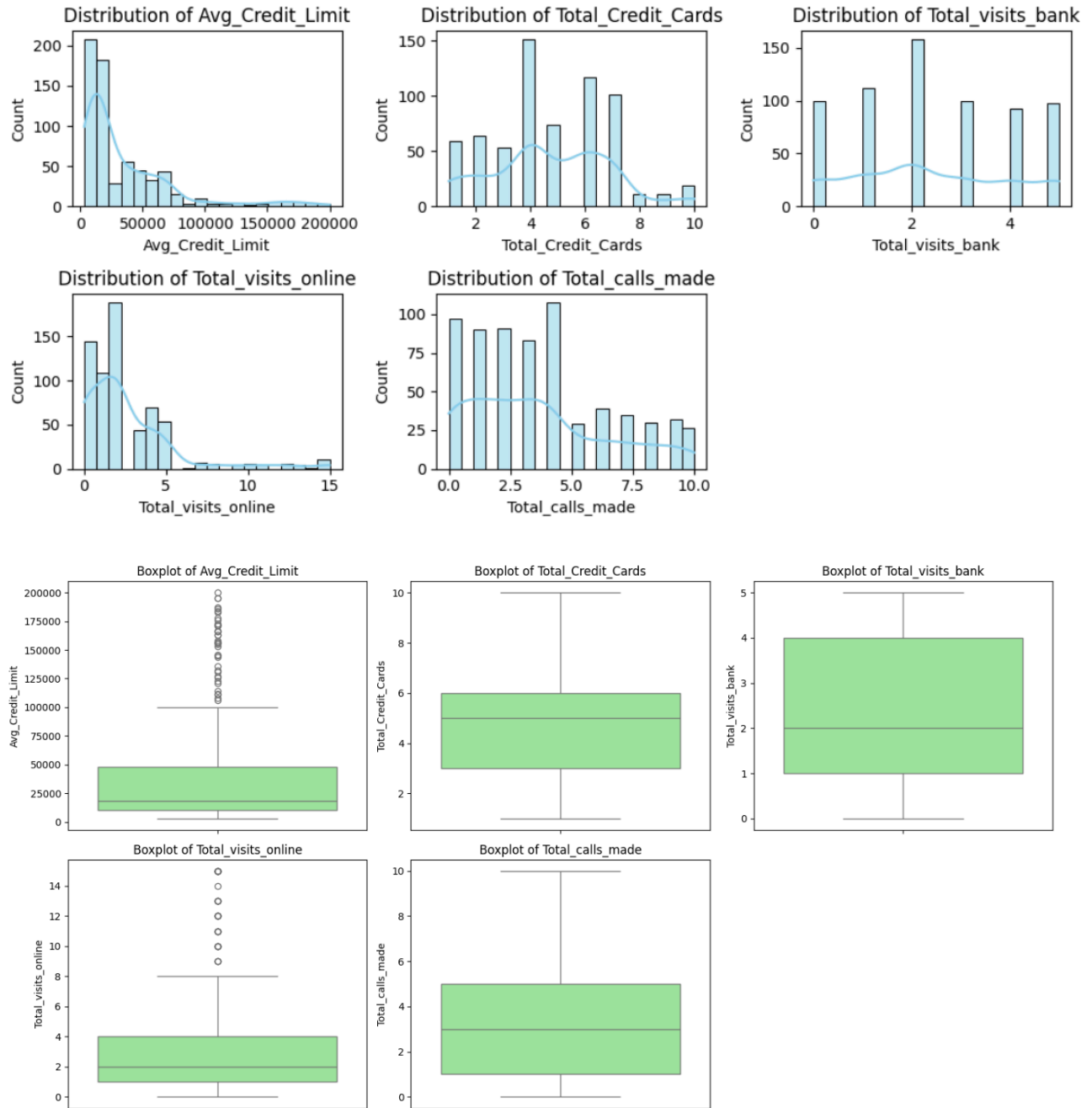
- Majority hardly visit banks.
- A smaller group still depends heavily on bank visits.

Total_visits_online:

- Distribution is skewed; some customers interact almost only online, while others don't at all.

Total_calls_made:

- A group of customers makes frequent calls, showing dependence on call centers.



Bivariate Analysis (Relationship Between Two Variables)

- **Credit Limit vs Credit Utilization:**

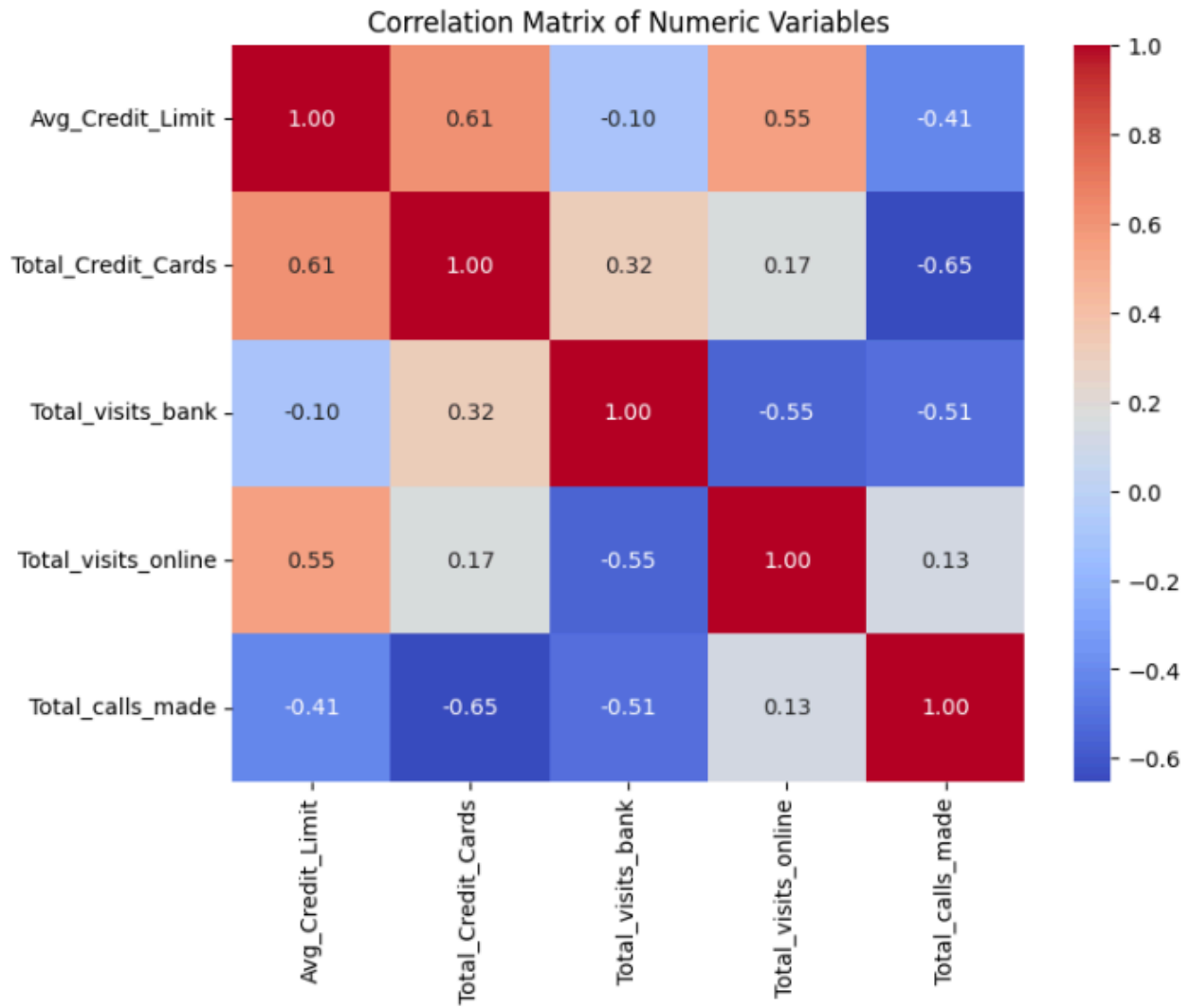
- Positive relationship; higher limits lead to higher utilization in absolute numbers.
- But low-limit users are seen to utilize proportionally more, meaning they might be under-served.

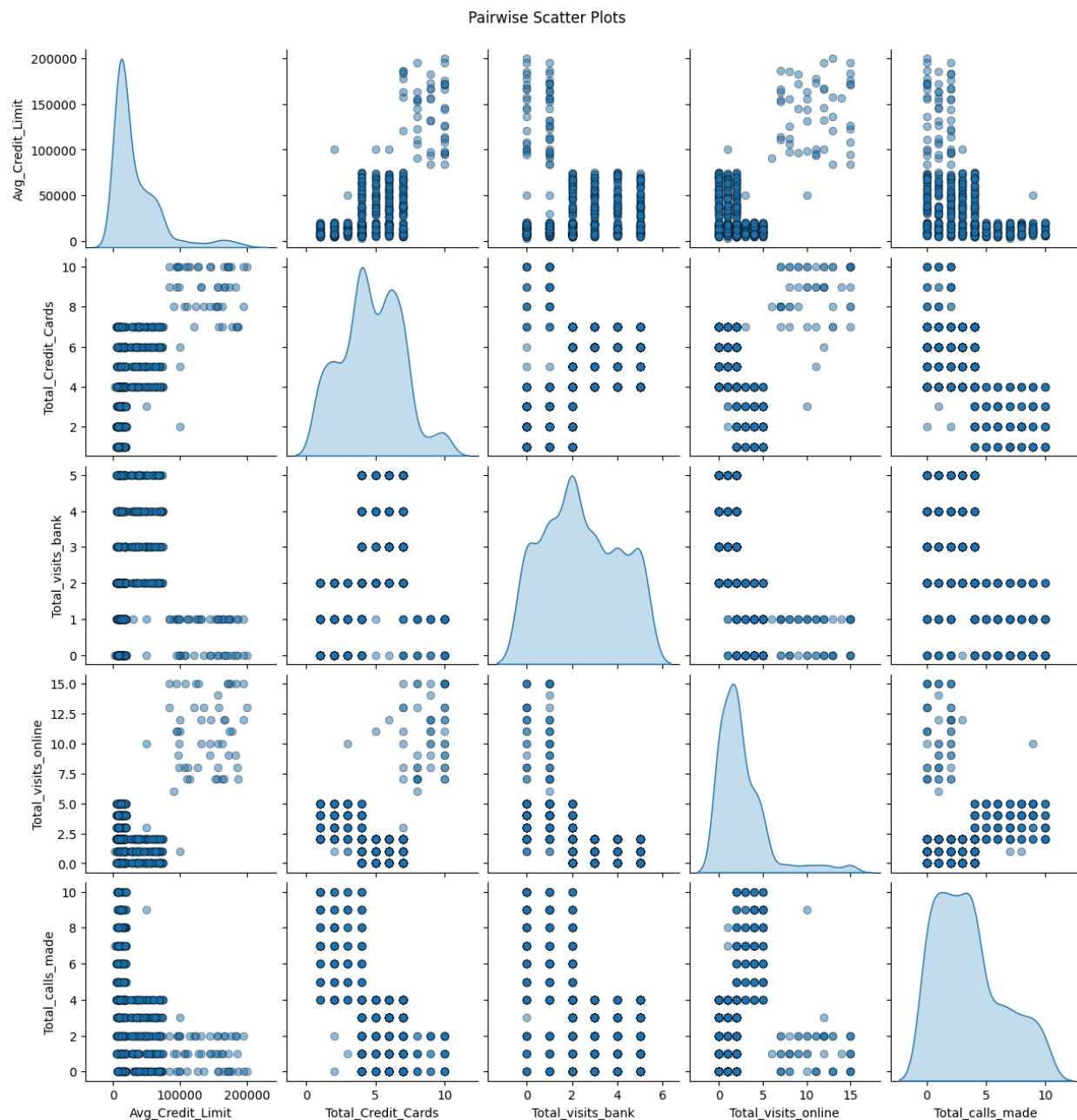
- **Online vs Offline Channels:**

- A clear trade-off exists — customers either go digital or remain offline. Rarely both.

- **Number of Cards vs Credit Limit:**

- More cards = higher limit. Indicates limit allocation depends on card holding.





Key Observations

1. Customer base is diverse: from digital elites to branch-dependent customers.
2. There are clear groups forming even before clustering.
3. Outliers exist in credit limits and utilization (treated in preprocessing).
4. Online and offline channels show opposite patterns.

3. Data Preprocessing

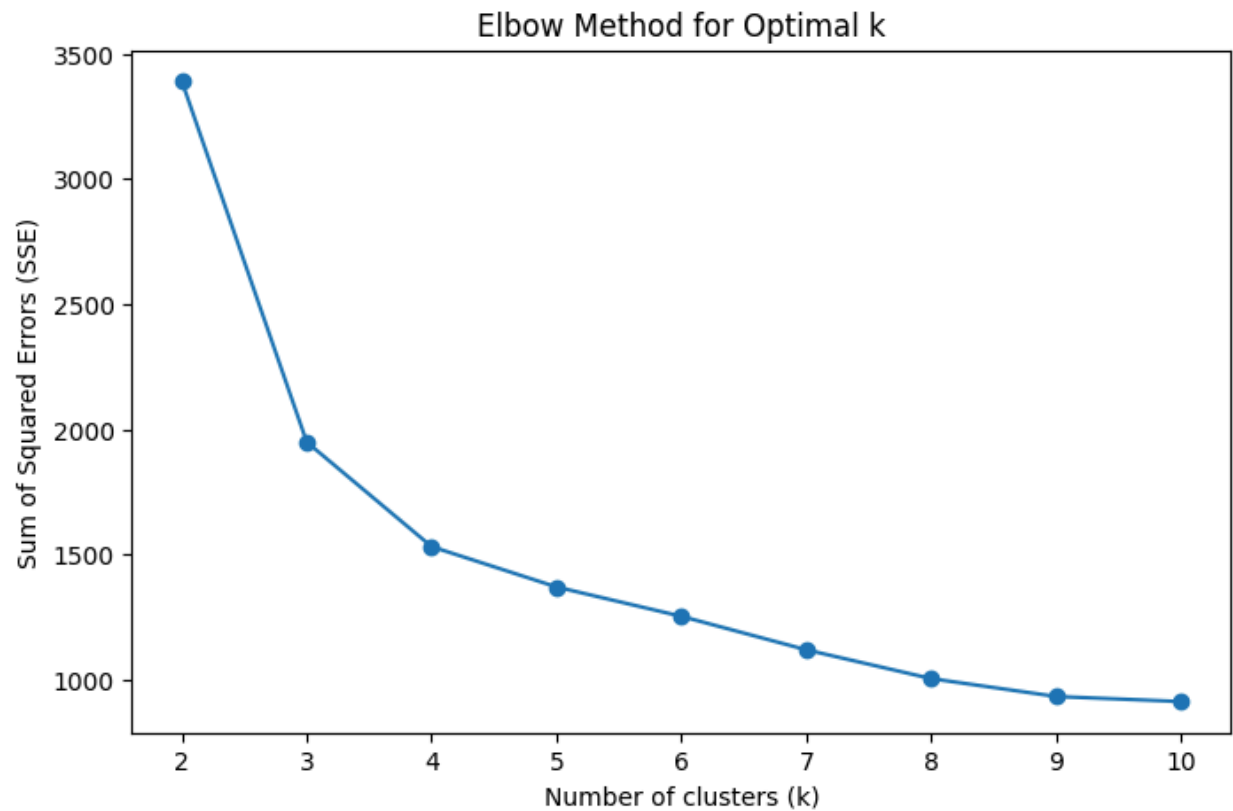
- **Missing Values:** Dataset had none. This ensured smooth clustering.
- **Outliers:** Handled using capping (winsorization) so extreme values do not distort clustering.
- **Feature Engineering:**
 - *Credit Utilization* = Avg Credit Limit ÷ Number of Cards → tells how much credit is being used per card.
 - *Bank Interaction* = Visits to bank + Calls made → tells reliance on offline support.
- **Scaling:** StandardScaler was applied so that all features (limits, visits, cards, etc.) are on the same scale. This is important because clustering is distance-based.

Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made	Credit_Utilization	Bank_Interaction	
0	2.398942	-1.249225	-0.860451	-0.619767	-1.251537	8.936352	-2.013705
1	0.643619	-0.787585	-1.473731	2.705813	1.891859	2.031639	1.217034
2	0.643619	1.058973	-0.860451	0.267054	0.145528	0.058864	-0.398335
3	-0.058511	0.135694	-0.860451	-0.619767	0.145528	-0.177869	-0.398335
4	2.398942	0.597334	-1.473731	2.705813	-0.203739	2.031639	-1.206020

4. K-means Clustering

Elbow Method

The Elbow plot showed that the optimal number of clusters is **3**.



Silhouette Score

Silhouette Score for $k=3$ = **0.427** → shows moderate clustering quality (acceptable separation between groups).

K-means Cluster Profiling

- **Cluster 0 – Mass Market Moderates**

- Avg_Credit_Limit $\approx$ 33K
- 5–6 cards
- Balanced channel usage (neither heavy online nor heavy offline).
- Represents middle-class customers, moderate value.

- **Cluster 1 – Elite Digital Users**

- Avg_Credit_Limit $\approx$ 102K
- 9 cards on average
- High online usage, very low bank/call dependence.
- Represents wealthy, digital-first customers who are profitable but expect convenience.

- **Cluster 2 – Low-limit Offline Dependents**

- Avg_Credit_Limit $\approx$ 12K
- ~2 cards
- Heavy reliance on calls and bank visits.
- Represents traditional customers who are high-maintenance but low value.

<i>KMeans_Cluster</i>	<i>Avg_Credit_Limit</i>	<i>Total_Credit_Cards</i>	<i>Total_visits_bank</i>	<i>Total_visits_online</i>	<i>Total_calls_made</i>	<i>Credit_Utilization</i>	<i>Bank_Interaction</i>
<i>0</i>	<i>33493.54</i>	<i>5.51</i>	<i>3.48</i>	<i>0.98</i>	<i>2.01</i>	<i>6324.52</i>	<i>5.49</i>
<i>1</i>	<i>102607.84</i>	<i>8.61</i>	<i>0.61</i>	<i>8.04</i>	<i>1.06</i>	<i>12791.99</i>	<i>1.67</i>
<i>2</i>	<i>12184.68</i>	<i>2.40</i>	<i>0.93</i>	<i>3.57</i>	<i>6.91</i>	<i>6426.80</i>	<i>7.84</i>

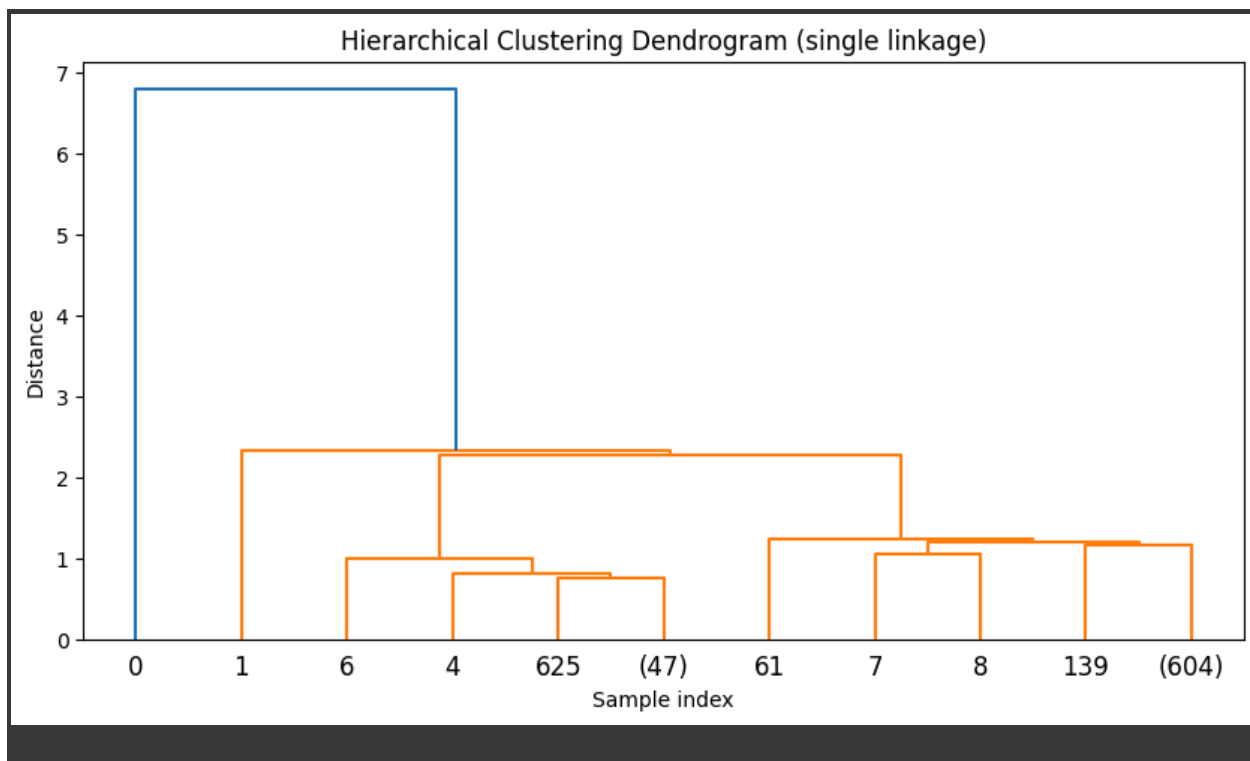
Hierarchical Clustering

Linkage Methods & Results

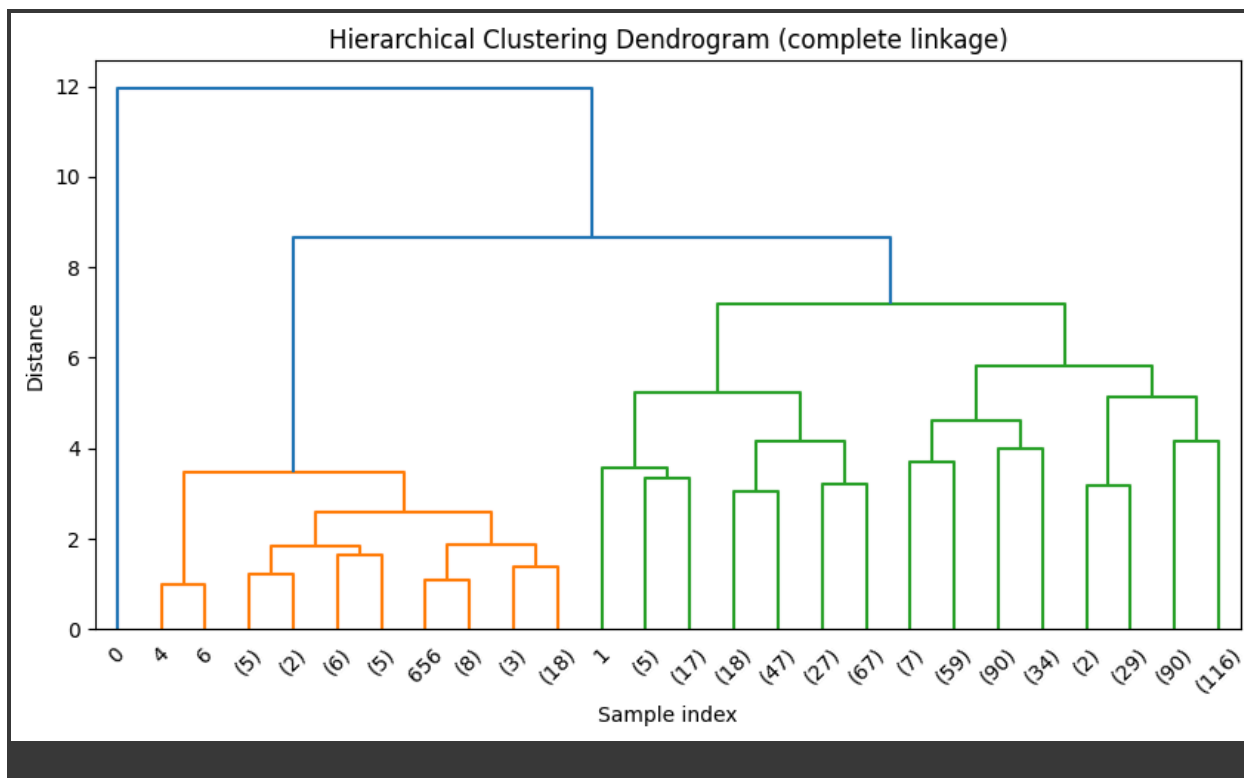
- Single linkage: Cophenetic = 0.671
- Complete linkage: Cophenetic = 0.794
- Average linkage: Cophenetic = 0.850 (best)
- Ward linkage: Cophenetic = 0.677

This means Average linkage gave the most reliable clustering structure.

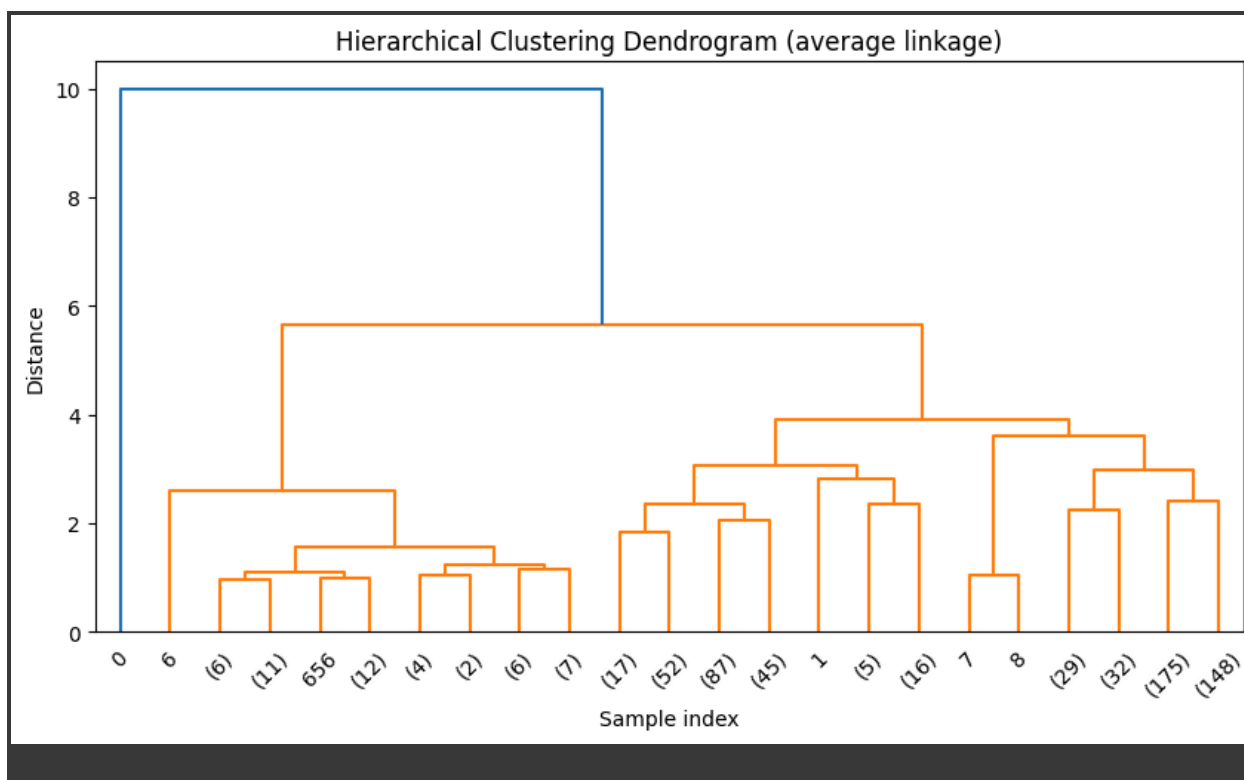
Cophenetic Correlation (single linkage): 0.671



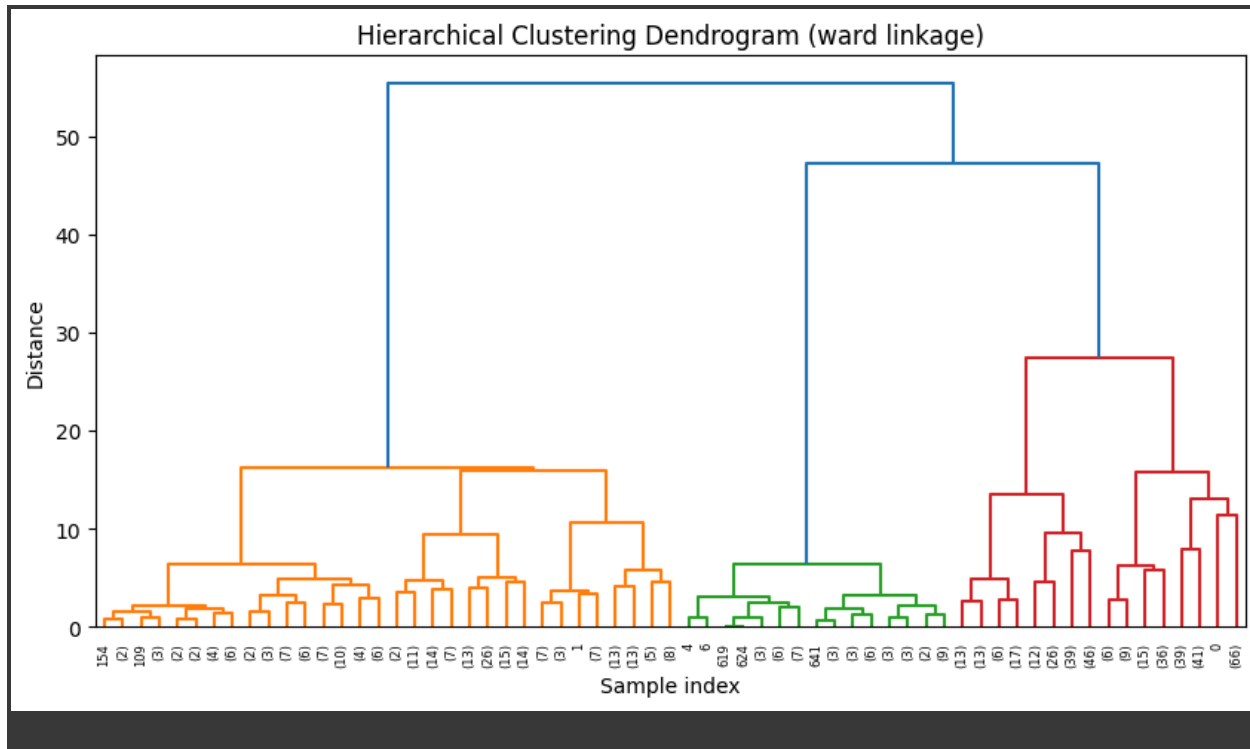
Cophenetic Correlation (complete linkage): 0.794



Cophenetic Correlation (average linkage): 0.850



Cophenetic Correlation (ward linkage): 0.677



Optimal Number of Clusters

Chosen = 3 (to match K-means and compare).

Hierarchical Cluster Profiling

- Cluster 0 – Elite Digital Users
 - Similar to K-means Cluster 1.
 - High credit limits, high card usage, online-focused.
- Cluster 1 – Moderates
 - Similar to K-means Cluster 0.
 - Balanced behavior.
- Cluster 2 – Dormant Elites
 - Avg_Credit_Limit $\approx$ 100K
 - Only 2 cards
 - Very low activity (hardly any visits or calls).
 - Unique group not captured by K-means.

Comparison: K-means vs Hierarchical

Aspect	K-means (3 clusters)	Hierarchical (3 clusters)
Overlap	Both identified Digital Elites & Moderates	Same groups identified
Difference	Identified Offline Dependents	Identified Dormant Elites
Stability	More scalable for large data	More detailed, shows hidden structure
Business Value	Good for segmentation at scale	Helps discover niche groups

Both methods are useful. K-means is practical, but Hierarchical gives richer insights.

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Actionable Insights & Recommendations

1. Elite Digital Users

- Offer premium credit cards, reward points, travel perks.
- Keep improving digital app experience.
- Upsell wealth management services.

2. Mass Market Moderates

- Introduce cashback and bundled offers.
- Encourage digital channel adoption through incentives.
- These customers can grow into elites with the right nudges.

3. Low-limit Offline Dependents (K-means only)

- High servicing cost.
- Encourage them to shift to online platforms with training and offers.
- Consider credit education programs.

4. Dormant Elites (Hierarchical only)

- High potential but under-utilizing credit.
- Launch reactivation campaigns (personalized offers, exclusive upgrades).
- Engage with targeted communication to revive usage.

Conclusion

The analysis successfully divided customers into 3–4 meaningful groups.

- The Digital Elites are high-value, tech-savvy, and profitable.
- The Moderates represent the mass-market backbone.
- The Offline Dependents are costly to maintain but can be nudged digital.
- The Dormant Elites (only visible in Hierarchical clustering) represent hidden opportunities for growth.

This segmentation allows the company to personalize strategy, cut costs, and maximize profitability.

END